

Yashvardhan Gupta

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SUMMARY

MS AI researcher with a focus on generative modeling, world models, and video synthesis. Hands-on experience implementing flow matching dynamics models, discrete diffusion language models, and multimodal video pipelines with measurable results. Background in reinforcement learning and sequential decision-making, relevant to perception-action loops in physical intelligence systems. Targeting research and engineering roles at the intersection of generative AI and robotics.

TECHNICAL KNOWLEDGE

Languages : Python, Java, SQL, Bash

ML & DL Frameworks : Jax/Flax, PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV

Systems & Devops : Docker, FastAPI, Flask, CI/CD, GCP, AWS, CUDA, ONNX Runtime, Git

Generative Modelling Flow Matching(ODE/SDE), Discrete Diffusion, VQ-VAE, Transformers, RAG, LangChain

Video & Vision U-Net dynamics models, video world models, multimodal pipelines, neural rendering (nerfstudio, gsplat)

EDUCATION

MS in Artificial Intelligence, Northeastern University (GPA: 4.0/4.0)

Sep '25 — Present

- **Relevant Coursework :** Machine Learning, Foundations of Generative AI, Reinforcement Learning

B.Tech in Mechanical Engineering, Delhi Technological University (GPA: 3.3/4.0)

Aug '19 — Jul '23

- **Core Courses :** Computer Vision, Machine Learning, Engineering Economics

PROJECTS

MicroDLM – Discrete Diffusion Language Model, Northeastern University [Link](#)

Feb '26 — Present

- Implemented a **10.7M-parameter discrete diffusion language model** from scratch in PyTorch — **bidirectional transformer (6L/6H/384E)** with **RoPE**, **RMSNorm**, **ReluSquared activations**, and **QK-normalization** — trained on **Tiny Shakespeare** via **absorbing-state masking** with a **cosine noise schedule**.
- Derived and implemented **SUBS parameterization** (clamping the MASK logit to $-\infty$ during generation) and **confidence-ranked parallel decoding**, achieving **$\sim 6\times$ fewer forward passes** than autoregressive generation (39 vs. 225 steps for a 240-token block).

MiniGenie - Flow Matching Video World Model, Northeastern University [Link](#)

Jan '26 — Feb '26

- Built a **42M-parameter pixel-space flow matching U-Net** in **PyTorch** to predict next video frames **conditioned on discrete player actions**, trained on **5,000** procedurally generated CoinRun (Procgen) episodes over **80K steps**.
- **Designed** Adaptive Group Normalization (**AdaGN**) conditioning to inject action embeddings and flow timestep signals into every ResBlock; used 15-step **Euler ODE** with **classifier-free guidance** and 10% **CFG dropout** during training.
- Trained a **6.1M-parameter VQ-VAE tokenizer** with a **512-entry codebook** achieving **100% codebook utilization** and **31.12 dB PSNR**; the full dynamics model reached 26.75 dB single-step PSNR on held-out rollouts.

Read my lips (Visual Speech-to-Avatar for Speech Impaired People), Northeastern University [Link](#)

Sep '25 — Dec '25

- Built a multimodal pipeline converting silent lip movements $\rightarrow$ synthesized speech $\rightarrow$ synchronized avatars.
- Integrated Auto-AVSR, Qwen 0.6B (semantic correction), and FLOAT (flow matching) for end-to-end generation.
- Engineered the system for **Apple Silicon (MPS)** by porting CUDA-centric generative models and optimizing tensor operations, achieving **$\sim 200\text{ms}$ VSR inference latency while maintaining high-fidelity output for real-time applications**.

WORK EXPERIENCE

Khoury College Teaching Assistant - Reinforcement Learning & Sequential Decision Making

Jan '26 — Apr '26

Northeastern University

San Jose, United States

- Guided graduate students in implementing and debugging core reinforcement learning algorithms (**DQN, PPO, Policy Gradients**), supporting convergence analysis, reward shaping, and training stability in PyTorch-based experiments.
- Conducted recitations and office hours on Markov Decision Processes (MDPs), **exploration-exploitation tradeoffs**, and **actor-critic methods**, providing structured feedback on experiment design, reproducibility, and **hyperparameter tuning**.

Digital Solutions & Technology Engineer

Apr '23 — Apr '25

Biowolk Healthcare

Delhi, India

- Engineered a scalable analytics microservice using **Python (FastAPI)** and **Docker** to process daily sales and inventory data, effectively reducing manual reporting time by **15% (~ 7 hours/week)** via **end-to-end data pipeline orchestration**
- Implemented automated budget-allocation and prediction pipelines (Python), reducing monthly spend variance by 8% through batch retraining and CI/CD model rollout.

PUBLICATIONS

Vision Language Models : A complete survey (Ongoing)

Dec '25

Reviews Vision-Language-Action models that combine visual perception, language grounding, and action generation for robotics. The paper identifies strengths, gaps, and opportunities for building next-generation embodied systems.

An End to End Solution to Automated Hiring

Dec '22

Proposed and evaluated a GAN-, NLP-, and CV-driven e-recruitment platform automating resume short-listing, deepfake-simulated interviews with dynamic question generation, and CV-based proctoring to accelerate hiring.
<https://ieeexplore.ieee.org/document/10060436>